

Introduction:

This Risk Assessment reviews the Stashi household network. Out of an inventory of 98 items, there were six critical items chosen to perform a full risk assessment on – one for each element. The first item selected was the people element PE – 011 Jane Stashi, or me. I like to think of myself as a low-risk person; however, that would not be completely honest statement. I am a student, an employee, a wife, a daughter, a sister, a friend, a niece, and ultimately a human. This means that I am inherently risky. I have family spread across the US, on the east coast, in Houston, Austin, Amarillo, and San Antonio. I used to have many hobbies such as playing tennis, reading, working out, cooking, all of which have been put on hold while I am still in school these last few semesters due to the difficulty of the classes increasing and the difficulty of work becoming two things that encompassed all of my time. I would say some of my riskiest behaviour stems from the extreme burnout and fatigue that I have experienced. Both factors have been a direct cause of a continued pattern of unhealthy behaviour and physical injuries. The next element selected was the networking element NW – 001 Eero Pro E6, the router. This router was provided by Vexus, our chosen internet provider, and was manufactured by Amazon. Amazon as the manufacturer provides any software updates that will assist with keeping up with security measures; however, our network availability is dependent on Vexus providing us the service we pay for. The next element selected was the hardware element, to which HW – 019 MacBook Pro was chosen. This is my personal laptop, which I use to do many things, such as perform bookkeeping activities, perform homework, research various topics, plan trips, access workout videos on, order HelloFresh meals, shop, play games, and perform many other functions. The risk behind the MacBook is that it is getting old for an electronic and is running out of storage space. The next element selected was the software element, to which I have chosen SW – 005 iCloud, which is used for some data storage including back-up storage on several of our devices. The risk that is included with this is that should we ever stop paying for iCloud services for a prolonged period, that data will eventually get deleted. The data element selected was the DT – 008 Gun Safe that we have, which is Rhino Metalworks gun safe, which are very high quality and secure safes, that we use to store not only guns and ammo but also important records such as birth certificates, our marriage certificate, social security cards, passports, and other important records. This gun safe is very difficult to break into and very heavy, so even if someone were to break into the house, they would not be able to easily remove the safe from the house. Furthermore, it would take several hours and extensive tools to drill into the safe. The safe is also fireproof and can withstand extreme temperatures for an extended period. The last element selected was the procedure element, to which PR – 009 Bookkeeping Routine was chosen. This is the routine that I perform to keep track of our finances. The program used for this is Quicken, which pulls the transactions directly from the connected bank accounts. The risk associated with uploading financial information into

Quicken this way is that there may be errors in data uploaded, such as it being uploaded twice, not uploaded at all, or uploaded incorrectly. This concludes the list of elements and selected devices for the risk assessment.

Overview of Inventory Conduction:

The inventory for the Stashi Network/Household was taken using several methods. The first method used was to review previously written papers for other classes. I then ran the program NMAP to determine what assets are currently registered as connected to the network. Lastly, I took a physical inventory by walking around the house and writing down the assets that I saw. Once I had all assets written down, I sorted them by their respective elements. I have included the detailed process from part one of this assignment below.

“... generate the table log in excel. I did this by creating a process for assigning the asset ID labels and system ID for all the devices. To create the asset IDs, I identified the different information system elements created, which include people, procedures, data, software, hardware, and networking. For labelling people assets, the ids are all generated with starting “PE - #”. The procedural assets are all labelled as “PR - #”. The data assets are all labelled as “DT - #”. The software and hardware assets are all labelled as “SW - #” and “HW - #” respectively. Lastly, the networking assets are all labelled as “NW - #”. The system IDs are all “SN - #” which stands for “Stashi Network.”

After I created these processes, I started by identifying the hardware assets. I did this by pulling data from previous classes to analyse work done in previous courses that might be beneficial to review as a part of this assignment. One paper that was done that I had saved from CIDM6340 was a Ransomware Recovery paper, in which I had created lists that included “an inventory of network devices, servers that [would] need to be backed up, devices that [would] need updates, identify passwords that [would] also need to be backed up, and a prioritized list of components for recovery” (Aldakkour, 2024). Another assignment I was able to utilize was from this course – CIDM6341 – the Wireshark assignment. As a part of the research I performed for this assignment, I ran several network mapping scans to identify which assets were assigned which IP and MAC addresses, from which I generated a list of assets that are connected to the network. I used this information from this paper to generate a list of Internet of Things (IoT) devices within my personal network (Stashi, 2025). I then compared the current list to the list from the Ransomware Recovery paper to identify differences.

There were several devices that were listed on the Ransomware inventory list that were not registered through the NMAP scan performed for the Wireshark and Phishing Assignment inventory list. The differing items are the dell laptop (Xcel), Xbox, Alexa A (Living Room), Alexa C (bedroom), Alarm System, and Speaker (Living Room) from the Ransomware list which did not appear on the Wireshark list. I determined the reason for each

missing device. The dell laptop (Xcel) is missing from the network scan because I no longer work at Xcel and returned their company equipment on my last day. The Xbox was not on the scan, as it is powered off at the time of the scan; however, it is still a device that is considered a part of the network. Alexa A (Living Room) is currently unplugged and being stored in a shed in the back yard, which is why this Alexa is not appearing on the network scans. Alexa C (bedroom) is also currently unplugged; however, it is being stored in the bedroom. The Alarm system is currently turned off and not in use, hence why it is not connected to the network. Lastly, the living room speaker is connected to the TV physically through a HDMI cable. Since it is not using Wi-Fi to connect to the TV and rather is using a physical cable to do so which means that it does not have a need to connect to the internet to send signals. With this comparison complete, I then went through and added devices from the Ransomware list that needed to be accounted for in the current asset inventory in the section(s) it was appropriate to add them. Following this, I then went through each room in the house and identified all hardware devices, to which I found a few more to add to the list. One device is my new work laptop, which connects to the network when I telework. Some others included speakers which use Bluetooth to connect to other devices, my camera which has Wi-Fi connectivity capabilities, and a digital photo frame that connects to Wi-Fi to upload new photos.

The next section of assets that I generate a list for were software assets. To start, I used the list of hardware devices to determine software such as operating systems that would be used by such devices. I then generalized what software would be used by a network or by all devices on a network to communicate, for security measures, and naming devices. Once these were listed, I then went to my iPhone and used the "Screen Time" feature on my systems setting application to determine which apps I use the most and looked at the data from the previous week to do this. I then listed the top 15 apps that were listed. The next section of assets that I took inventory of were the people assets. I took inventory of this section by referring to various legal documents, such as birth certificates and marriage licenses. I also utilized the family tree records on Ancestry.com to locate documented records of familial connections within my network (this family tree was verified through genetic testing). Ancestry.com includes both mine and my husband's family trees. To determine documented "friends" within the network I referred to my contacts list within my phone, for identifying co-workers, I can access the company directory located on the company's employee site, and to identify acquaintances and distant friends I utilized social media sites such as Facebook and Instagram. There were several ways in which I determined the procedures section of this assignment. The first was to review my planner, which is where I journal daily. In this journal, there are large sections for each day of the week by week and a section in the front that includes a monthly calendar for each month. In the monthly section, I include bigger events, such as birthdays, parties, trips, etc. In the week section, I write a summary of what I did each day. This often includes a short summary of routines, what I did each day, and any strong feelings that I experienced. Using this was helpful in not only identifying certain routines that I had, but also in seeing the patterns when routines certain routines were being practiced and when others were not. For example, some routines that are

noted when they are being performed are when I am in the habit of cooking or working out. When I am not in the routine of doing these items, I document them well even including the time I started the activity, details about it (such as the meal cooked if cooking, or the workout performed). Another way I gathered data about routines was by going through my workday. I have routines that I do before going to work and after leaving work that are in a different category of compared to the rest of my day outside of work due to the secure nature of the it. The last section completed was the data section. This section was completed by first identifying all repositories that I have. Once these repositories were identified, they were then reviewed for the type of data being stored within them.” (Stashi, 2025).

Overview of Asset Categorization:

Each asset was categorized based off which element it best fit under. There are several assets that fit under several different element types, and as such, they were listed several times under each element. For example, several devices were listed as both networking devices due to their functions and hardware devices due to their physical presence. From the full listing of assets inventory, I selected what I determined to be the most critical asset from each element to be representative of that asset throughout the risk assessment. I determined these assets to be the most critical through the overall effect they would have on other network assets should they be made unavailable.

Overview of Threats/Vulnerability Identification:

The next section that I worked through was the threat categorization and vulnerability description sections. The threat categorization section is broken into three sections, “confidentiality,” “integrity,” and “availability.” The confidentiality section focuses on ensuring information is not given to unauthorized individuals. The integrity section focuses on ensuring that there is not unauthorized changes or destruction of data. The availability section ensures that there is timely and reliable access to information at any given point. A rating of high means that there is expected to be a catastrophic adverse effect to the network and/or individuals, a rating of moderate means that there is expected to be a serious adverse effect, and a rating of low means that there is minor adverse effect. PE – 011 Jane Stashi had a rating of C: High, I: High, and A: High. This was determined, as I know where information is, the passwords, and how to gain access to every account, safe, and other places information is secured. Furthermore, there are many processes that I perform as well. If I were to be compromised it would have catastrophic consequences for the network, as many processes would not be completed and much knowledge would be lost. Vulnerabilities for confidentiality were intoxication, as this is something that could lead to my defences being down and that would lead to me releasing information to users who should not have it. Furthermore, other vulnerabilities include having the human factor, as I can fall for social engineering tricks that would give up more information to unauthorized users who do not or should not have it. Integrity is rated high as well, as it is critical that I remember things such

as passwords and processes. Vulnerabilities here are amnesia, time, and a single point of failure. Over time, my memory will change, and I will forget things. Furthermore, there is a single point of failure, as only I know certain passwords and processes. This affects both availability and integrity of data. Lastly, availability is rated high. If I cannot be accessed, then there is a lot of risk as earlier stated I am the only one who knows how to access certain items or how to perform certain processes. The availability of myself is vulnerable to amnesia, death, time, and single point of failure again.

The next element was the networking element of Eero pro E6 Router NW – 001. The router was assigned the values of C: High, I: High, and A: High. I determined the confidentiality was high because the router connects to many network devices, and therefore it was important that only authorized users be able to access router without it causing severe consequences. I considered the integrity to be high, because although there would be severe consequences to the network should the data that the router be communicating be inaccurate (or if the data's integrity is compromised). This is because of the router's communications to network devices and external ones. Should the router have the wrong IP address to send a packet through to, it would start causing immediate issues within the network. Availability was also rated high because without the router being able to function, the system will see major negative impacts immediately as many devices such as the TVs, and other devices that connect to the internet will be rendered useless or impaired immediately. Vulnerabilities to the Eero pro E6 were determined through researching and analysing the network to determine where the vulnerabilities call. For confidentiality, bad users were likely to gain access to the network through weak passwords being used. Open ports were another vulnerability to the router. For integrity, the vulnerability lies in bad actors gaining access to the network and changing or erasing data. Lastly, the vulnerability for unpatched firmware, as patches usually provide necessary security updates.

The next element was the HW – 019 MacBook Pro, which is the hardware device chosen for the risk assessment. This device was assigned the values of C: High, I: High, and A: Moderate. This was determined, as there is much data on my laptop that must maintain its integrity and that would be detrimental to the network should unauthorized users gain access to it, which is how the confidentiality and integrity aspects were determined. The availability is rated as moderate because the data is backed up to iCloud, and therefore, the hardware of the laptop itself is not mandatory for the functionality of the network. The laptop's hardware can be replaced and the data restored from the cloud. The vulnerabilities if this element were determined as such: confidentiality was determined as someone unauthorized could get in through a weak password, integrity was determined as missing endpoint protection or antivirus protection as malware could erase or change vital data, and availability vulnerability was determined as the laptop being lost by my own human error or stolen.

The software element chosen was SW – 005 iCloud, which was rated high in all three CIA categories. It was rated highly in all categories because the iCloud software is vital to business operations. Devices connect to the cloud every day to pull data from it as well as add data to its storage and back-up data to it. Because it contains such important data, it is

extremely important that it remains confidential and no unauthorized users gain access, that the data maintains its integrity, and that the cloud is readily available to access.

The data element chosen was DT – 006 Gun Safe, which was rated high on confidentiality, but low in integrity and availability. It was rated highly in confidentiality because of the nature of the data stored. Should bad actors gain access to this data - which includes social security cards, birth certificates, etc – they can commit to identity theft and other crimes related. Integrity and Availability were rated low on this scale, however, as the integrity of these documents would be very difficult to change without my knowledge. Furthermore, it can be verified with the government’s records to verify any correctness. Availability was rated low as there are very few times were not having access to these documents will be detrimental to the operations of the network. These records can also be replaced should they be lost, stolen, or destroyed. The threats that are faced by the data element are unsecured access to the safe by it being left open, as someone can then take whatever they like from the safe should they so choose. The lack of an audit log affects the integrity of the safe’s data, as there is no record on who accessed it at what time. The last vulnerability listed was the availability, which exists should there be a power outage or lack of electricity to power the safe. The safe requires power to be locked or unlocked, and so without it the safe cannot be opened or closed.

The last element to discuss is the procedure element, to which PR – 009 Bookkeeping Routine was selected. The CIA was rated as moderate across the board. This was determined as this process is not vital for the network’s functionality, and while the data is sensitive it is not detrimental to the network should it get out. It is not ideal for it to be shown to unauthorized users. The vulnerabilities that this element faces are that there is a lack of segregation of duties, which is a well-known fraud risk. Since this is Austin and I’s personal bank accounts this is a very low risk, as my money is my money and his money is also my money (only in spirit! Haha! He has access to see all bank accounts and can verify where money is going should he so choose to). There is also the vulnerability of data being uploaded into the financial software incorrectly.

Overview of Asset Prioritization and Value:

The next part of this section was to assign a priority to each of these assets. The priority ranged from Vital - business cannot operate if lost more than a day, Critical - business cannot operate if lost 2-5 days , High - major disruptions if lost more than a week, impacts parts of the organization immediately, Medium - major disruptions if lost more than a month, impacts parts of the organization immediately, Low - impacts parts of the organization, Facilitates - minimal impact on organization, impacts individuals. I rated each item based off of these ratings. For PE – 011, myself, I rated it as “high” because the home organization would be able to last about a week before being seriously impacted. This means that several routines not being performed, while some other parts would be impacted immediately, such as Austin. The networking element – the Eero Router – was rated as being

“vital” to the network. This is because the network cannot operate more than one day with it lost. The router connects all devices to the internet. Without this connection, many of the devices would not be able to function at their full capacity, which would have a large impact on the network and hence it’s rating as “vital.” The hardware element, my MacBook, was rated “high” to the network. This device is something that is used on daily basis; however, the network could survive a few days before the effects of the absence of this device will start to truly be felt. The network can survive without this device; however, it would cause major disruptions in several processes, such as my bookkeeping process, homework process, and ordering HelloFresh to name a few. The software element, iCloud, is utilized to store data and to back-up data for several devices on the network as well. This makes it very important to be able to have access to this software and to keep it secure, which is why it was rated to be “vital” to the network. Furthermore, since data is being stored through iCloud, should connection or access to it be lost more than one day it would severely impact the network. The data element, the Gun Safe, is rated as having a “facilitates” impact on the organization. There are several important records stored in there, including birth certificates, social security cards, passports, and other identifying information. Losing this information has low impacts on the network, as it would impact parts of the organization – the people – however it is replaceable. Losing this information would have minimal impact on the people in the short term, however it staying missing in the long term would increase its effects on the people. Lastly, the bookkeeping routine was rated “facilitates” because it has minimal impact on the organization. This routine is performed to maintain and analyse financial records. I perform this routine to track our financial health and find ways to optimize it. This has minimal impact on the rest of the network's operations.

The values for each asset were thoroughly thought out and assigned. The first asset on the asset inventory list chosen for this risk assessment is the people system element. I chose to focus on myself as a people information system element. I have assigned a CIA value of \$150,000.00 to myself. I determined this value based upon several factors, which include a life insurance policy I have on myself through work, the cost of the education that I have invested into myself and added value of various other skills and knowledge that I have acquired. The second asset is a networking element, for which I chose the Eero Pro6 router as the asset, I have assigned a CIA value of \$15,000 to it. I have assigned it this value based on the device’s value, which is \$89.00 on Amazon (Eero, LLC, n.d.). I also assessed all the devices that would be rendered useless without the router, as well as the fact that a major part of the network would be inoperable. This value came out to be about \$15,000. The hardware device element chosen was my laptop, HW – 019 MacBook Pro. This device was assigned a CIA value of \$2,000. The hardware on the MacBook itself would cost \$2,000 to replace should the need arise (Apple, n.d.). I only considered the value of the hardware itself, not the data on the laptop. It was done this way since the data on the laptop is contained within a backup through iCloud. The software element chosen was SW – 005 iCloud, which stores a lot of data for the network including device back-ups which would be vital to recovering the network or a device from an attack. The assigned value was \$100,000, for the cost of recovering iCloud data, which can be up to \$1,000 per device for high quality recovery, the cost per month it would be to have 2T of data is \$11.99 per month, \$143.88 per year, or

\$7,913.40 for the next 55 years (Apple, 2025). The data element that was chosen was the gun safe, which stores important documentation such as birth certificates, our marriage license, social security cards, and other documents. The CIA value of \$5,000, which was determined by the cost of the safe (\$3,000) as well as the value assigned to the assets within (excluding the guns). The last element is the procedure element, to which PR – 009 Bookkeeping Routine was chosen. This routine was chosen to have a CIA value of \$1,000.00 that was calculated by determining the monthly payment for the software and the hours I put into analysing our budget (Quicken, n.d.). This demonstrates how the values for each element were selected.

Overview of System Determination:

The overall system determination of the asset PE – 011 Jane Stashi is that this asset possesses a high risk to the network should it be unavailable to the network. It was given an ARO score of “very high” or 0.20 based on the grading scale provided. This was determined as I took the rating of each CIA categorization – high, moderate, and low – and corresponded it to one of the provided ARO ratings. High corresponded to a rating of “very high” or 0.20, moderate corresponded to a rating of medium or .10, and low corresponded to a rating of very low, or .001. I then took the ARO rating score that corresponded with the score given for each CIA (confidentiality, integrity, and availability) area and averaged of them to determine the overall ARO score. So, for Jane Stashi, all CIA scores were high – or 0.2 - so the ARO score was determined to be 0.2 or very high. The vulnerabilities for each CIA area were then determined and addressed through controls which were established. For PE – 011 Jane Stashi, the vulnerabilities of intoxication, falling for social engineering, and being human were assigned. These are all vulnerabilities I have that put the confidentiality of the network at risk. For integrity, the vulnerabilities of time – as memory fades and changed as time passes – and amnesia exist for integrity. Lastly, the risk of amnesia, death, time and single point of failure exist for availability. The controls were then established for confidentiality, which were that I don't drink regularly as I am tested through work, and that I take trainings to keep up with recent scams/critically assess emails and messages from both known and unknown sources to control the human factor vulnerability and intoxication vulnerability. I scored myself at a 0.8 confidence level that these controls would work, as although I rarely drink, it still happens. Although I do my best to critically assess each email, I am still human and as seen in the phishing assignment, will be fooled by phishing attempts. Since my confidence level is 0.8, my uncertainty is 0.2. My risk value was then calculated through the formula provided, which turned out to be \$7,200.00. For the integrity and availability sections, I have addressed the vulnerabilities of amnesia, time, and single point of failure through the controls of providing documentation of processes and events through journalling, obtaining health insurance and life insurance. I have rated the certainty of the integrity section 0.6 because while I believe that the documentation will be 60% effective in mitigating the vulnerabilities. This leaves the uncertainty to be 0.4, and the risk value to be \$16,800.00. Lastly, the availability had similar controls in place such as the documentation processes, but I will also

try to avoid putting myself in dangerous situations as best as I can by practicing safe driving habits, avoiding risky parts of town, eating healthier, etc. I rated this working at a 0.4, as there are many things that are just out of our control. This left the uncertainty at 0.6 which made the calculated risk value \$28,800. This same practice was continued for each of the other elements.

For the asset NW – 001 Eero Router, the values of C – high, I – high, and A – high were assigned, which average to an overall ARO value of 0.20, or very high. This was determined as the router acts as the primary gateway for network traffic. The vulnerabilities for each CIA category were then determined and addressed through controls. The vulnerabilities for confidentiality were determined to be a default or weak administrative passwords, open ports, and wireless eavesdropping. These are things that could make it easy for a bad actor to gain access to a network. The controls established for these vulnerabilities are that Austin and I have worked to create a strong password that has numbers, upper- and lower-case characters, as well as special characters and does not have any commonly used words. Furthermore, I have the ability to run open NMAP scans to ensure that there are no open ports. The router also includes a built-in firewall. I rated my certainty that these controls would work at 0.80, as although the NMAP scan confirms the ports are closed and the firewall works to protect against outside presentations, I am still susceptible to social engineering attacks on Eero credentials and a brute force attack can still be used. Since my confidence level is 0.80, there is 0.20 uncertainty. For integrity, the vulnerabilities of bad users gaining access could change or erase data through DNS hijacking was assigned. These vulnerabilities were determined to be bad users gaining access could change or erase data through the DNS hijacking. This vulnerability is addressed through the control of the built-in firewall, which assists with security practices and blocks unauthorized users. I rated the certainty of the integrity section at 0.90, as a built-in firewall is an effective preventive control that will stop most threats, but not all. This leaves the uncertainty to be 0.10. For availability, the vulnerabilities of unpatched firmware and hardware failure exist. The controls in place for this are similar to the controls for integrity, such as the Eero router gets regular updates from the manufacturer (Amazon). I rated this working at 0.95, as the automatic updates are highly reliable and reduce the likelihood of weaknesses in the firewall. This left uncertainty at 0.05. The final risk value calculations are included in the asset management excel table.

For the asset HW – 019 MacBook Pro, the values of C – high, I – high, and A – moderate were assigned. These had an average score of 0.16667, which I rated as an overall ARO value of high (0.15). This was determined as the laptop is a primary asset used for work and personal data, meaning any compromise poses a major disruption to the personal organization immediately. The vulnerabilities for each CIA category were then determined and addressed through controls. The vulnerability for Confidentiality was determined to be default or weak administrative password. The controls established for this vulnerability are that Austin and I have worked to create a strong password that has numbers, upper- and lower-case characters, as well as special characters and does not have any commonly used words. I rated my certainty that this control would work at 0.80, as while the password is

strong, there's always a risk of exposure through observation or keylogging. Since my confidence level is 0.80, there is 0.20 uncertainty. For Integrity, the vulnerability of missing endpoint protection (antivirus) was assigned. This vulnerability is addressed through the control that Apple has preliminary measures such as Xprotect, Gatekeeper, Malware Removal Tool, as well as background security improvements to assist with protecting against viruses and malware. I have rated the certainty of the Integrity section at 0.80, as Apple's built-in controls are robust, but they do not provide the comprehensive, real-time scanning of dedicated third-party antivirus software. This leaves the uncertainty to be 0.20. Lastly, for Availability, the vulnerabilities of being stolen or lost exist. The control in place for this is that I keep it inside the house; if it leaves it always stays on my person; all back-up data is stored in the cloud. I rated this working at 0.90, as keeping the physical asset secured and ensuring cloud backups are current significantly reduces the likelihood of unrecoverable data loss or long-term unavailability. This left uncertainty at 0.10. The final risk value calculations are included in the asset management excel table.

For the asset SW – 005 iCloud, the values of C – high, I – high, and A – high were assigned, which average to an overall ARO value of 0.20, or very high. These average to an overall ARO value of 0.20, or very high. This was determined as the iCloud is used to back up device network data, meaning that if compromised or unavailable, the business (personal organization) cannot operate for more than a day. The vulnerabilities for each CIA category were then determined and addressed through controls. The vulnerability for Confidentiality was determined to be default or weak administrative passwords. This is a vulnerability as weaker passwords are more easily susceptible to blunt force attacks. The controls established for this vulnerability are that Austin and I have worked to create a strong password that has numbers, upper- and lower-case characters, as well as special characters and does not have any commonly used words. I rated my certainty that this control would work at 0.80, as strong passwords can still be compromised via phishing or data breaches. Since my confidence level is 0.80, there is 0.20 uncertainty. For Integrity, the vulnerability of shared account credentials was assigned. This is a vulnerability as I am not the only user on the iCloud account, which increases the area for attack more. This vulnerability is addressed through the control to mandate strong passwords for all users. I have rated the certainty of the integrity section at 0.80, as mandating strong passwords is a robust measure, but the integrity of the data still relies on the user behaviour of all account members. This leaves the uncertainty to be 0.20, as brute force attacks if given enough time, will eventually be able to determine the password. Lastly, for availability, the vulnerability of single point of failure in back-up data and the need to pay to maintain access exists. The control in place for this is to store a copy of the back-up file on a local disk or somewhere else. I rated this working at 0.80, as maintaining a local, independent copy significantly reduces the risk of long-term unavailability caused by cloud service failure or non-payment. This left uncertainty at 0.20, as the back-up saved to a local disk or somewhere else is not updated daily like back-ups to the Cloud can be. The final risk value calculations are included in the asset management excel table.

For the asset DT – 006 Gun Safe, the values of C – high, I – low, and A – low were assigned. These held an average of 0.067, to which I assigned the ARO value of low (0.05). This was determined as the asset contains important records, such as birth certificates, social security cards, and marriage certificates. These documents, while having a high impact if unauthorized access is gained, have a mitigated risk in the integrity and availability sections as these documents are also maintained by governmental agencies. Copies can always be requested, and permanent changes to these documentations will be extremely difficult to be made. Furthermore, the company I work for offers an add on to purchase legal services in our benefits, which I have. Part of the legal services offered are to assist in recovery in the circumstances that your identity is stolen. The vulnerabilities for each CIA category were then determined and addressed through controls. The vulnerability for confidentiality was determined to be unsecured access/remaining unlocked. This is a vulnerability as unauthorized persons could gain access to highly sensitive identity documents or valuables. The controls established for this vulnerability are that the locking mechanism will only open if powered on and the correct password is input; the safe's 900 lb weight and 6ft height make moving it difficult, and it would take a long time to drill into the side of it; and the safe is closed and locked any time it is not in use. The safe is also located within Austin and I's house. The house is surrounded by Ring security cameras, and all entry points are locked and secured. I rated my certainty that this control would work at 0.95, as the combination of physical security (weight/drilling difficulty) and strict procedural controls (password/locking) is extremely effective. Since my confidence level is 0.95, there is 0.05 uncertainty. The 0.05 uncertainty is because nothing is 100% effective and if given enough time and resources, someone could get into the safe. For Integrity, there is no audit log of access, which creates a vulnerability. This is a vulnerability because if contents were tampered with or removed, there would be no electronic record of who accessed the safe and when. This vulnerability is addressed through the control that Austin and I are the only ones who know the password to get in, so if anything goes missing or is removed from the safe it could only be one of us. I have rated the certainty of the integrity section at 0.90, as limiting knowledge of the password to two trusted individuals establishes a strong chain of custody, effectively replacing an audit log. This leaves the uncertainty to be 0.10. Lastly, for availability, forgetting a code, an electronic failure, or electric outages all are vulnerabilities that affect the availability of obtaining the contents of the safe. Should the contents be stolen or destroyed, this would affect the people in the business operations most; however, it is replaceable, as government agencies are also tracking this data metrics and can provide copies when needed. The control in place for this is to pay electric bills on time (to prevent power loss) and the safe's design, which provides a key override or backup battery. Furthermore, having a back-up generator would mitigate this risk as well. I rated this working at 0.95, as the electric outages are rare, and alternative methods for accessing the contents are available, such as a back-up generator. This left uncertainty at 0.05. The final risk value calculations are included in the asset management excel table.

For the asset PR – 009 Bookkeeping process, the values of C – moderate, I – moderate, and A – moderate were assigned. These average to an overall ARO value of medium, or 0.10. This was determined as the process is not vital for the network's

functionality, and while the data is sensitive it is not detrimental to the network should it be released to unauthorized users. The software used for tracking this, Quicken, is not set up with the permissions to access the bank accounts to move money itself. The vulnerabilities for each CIA category were then determined and addressed through controls which were established. The vulnerability for confidentiality was determined to be a lack of segregation of duties. This is a vulnerability because a single person (myself) handles all aspects of the financial data, increasing fraud risk. The control established is that Austin has access to all the bank accounts to verify their amounts and ensure money is not missing; he also has access to review bank statements. I rated my certainty that this control would work at 0.80, as the risk is mitigated by the small, high-trust circle of the personal organization. Since my confidence level is 0.80, my uncertainty is 0.20. I rated this control as 0.80 because Austin occasionally checks the accounts, however, he heavily relies on and trusts me to perform this process. For integrity, the vulnerability of manual data entry errors/upload entry errors was assigned. This vulnerability is addressed through the control to reconcile Quicken updates against bank statements for each account to ensure accuracy. I have rated the certainty of the Integrity section at 0.95, as the reconciliation process is highly effective at identifying and correcting data discrepancies, leaving minimal margin for error. This leaves the uncertainty to be 0.05. The 0.05 represents transactions that cannot be identified through descriptions. Lastly, for availability, the vulnerability of incomplete transaction logging exists. The control in place is also to reconcile Quicken updates against bank statements for each account to ensure all transactions are captured. I rated this working at 0.95, as the bank statements provide a complete, non by passable record, ensuring the availability of the data set for analysis. This left uncertainty at 0.05. The final risk value calculations are included in the asset management excel table.

Overview of System Categorization:

I determined the assets within the Networking element category by using multiple different approaches to ensure a complete and current inventory. The networking element has overlapping assets with several different elements, specifically it and the hardware categories have several overlapping assets and were often analysed together, as many modern devices function as both physical hardware and network connection points. I used the information for the six elements provided in the PowerPoint for Chapter 6, table 6-1, to help categorize what belongs where (Whitman, 2018). The overall process began by reviewing two sources of historical and current technical data, which was explained under Overview of Inventory Conduction section in this paper. To summarize, I reviewed prior academic work, such as a Ransomware Recovery paper, which provided initial lists of network devices and recovery priorities. I then ran the NMAP program to scan the current network and identify all assets registered with a corresponding IP and MAC address, generating a list of currently connected Internet of Things (IoT) devices. I then reconciled the historical inventory list with the current NMAP scan results to identify discrepancies. This analysis involved determining the status of missing items. Devices absent from the current NMAP scan (like the Dell laptop or

some Alexa units) were removed or noted based on their status (that is, returned to a previous employer, powered off, or unplugged). Certain devices were analysed for their connectivity type. For example, the living room speaker was determined to be connected via HDMI cable – a physical hardware connection – rather than Wi-Fi, thus eliminating its need to be listed solely as a networking device. Finally, I conducted a physical walkthrough of the house to identify and add any newly acquired devices or those that had been missed by the scans. This added assets like my new work laptop, Bluetooth speakers, and a Wi-Fi-enabled digital photo frame. After identification, all assets were assigned an ID based on how they fit into all elemental categories within their functions of the organization. For example, all of the hardware devices were also listed as network devices. This was because all of the hardware devices connected to the network and were network devices as well. There were other elements such as the DNS naming service that operates on a physical device but is also a separate asset of the network that is essential to functionality. This example specifically was listed under the software and networking elements. All the devices were analysed to determine what their entire functions were and how many different elements they could be categorized under. This comprehensive process ensured the list was both historically informed and accurate to the current state of the Stashi household network.

Overview of Risk Posture:

The overall risk posture of the Stashi Network based off on the system and risk analysis performed on the six chosen assets is that the Stashi Network maintains a moderate-to-high risk posture. The highest areas of risk are associated with the people and core system dependencies. The analysis identified three vital assets which are PE – 011 Jane Stashi, the NW – 001 Eero Router, and SW – 005 iCloud. All these assets were rated very high and assigned an ARO value of (0.20). These items drive most of the risk, primarily due to the detrimental and immediate impact their loss would cause the organization. The largest calculated residual risks are attributed to the human factor (social engineering, amnesia, single point of failure) and the reliance on third-party availability (Vexus, Amazon/Apple cloud services). While the network is defended by strong technical controls on hardware and software (e.g., NMAP verification, strong passwords, and multi-layered backups), these technical defences are only effective where human-related uncertainty is low. Furthermore, the overall risk is mitigated by the low-impact categorization of physical/procedural assets like the DT – 006 Gun Safe and the PR – 009 Bookkeeping Routine, whose high impact risks in confidentiality are effectively transferred or minimized through controls such as legal identity theft services and financial reconciliation. At this moment in time, it is the opinion of myself that the current risk, while rated high, is acceptable. I have trust that our personal information (social security cards, birth certificates, etc.) are very well protected in the safe, and that I have had sufficient training to avoid giving up data by falling for phishing scams. I also trust the services being provided from Vexus (for the Eero Router) and Apple (iCloud) and have yet to see a reason not to trust them. Both of these companies are well known, reputable companies. Moving forward, should I want to reduce our risk the priority is to

reduce reliance on PE – 011 by providing cross-training on critical processes and strengthening human-based authentication to lower the high uncertainty scores associated with personal and procedural risk. Should Vexus or Apple provide reasons to not be trusted, such as a decline in availability due to failure to services, I can always work to find new companies to provide these services.

References:

- [1] Eero, LLC. (n.d.). *eero 6 dual-band mesh Wi-Fi 6 router*. Amazon. Retrieved November 8, 2025, from <https://www.amazon.com/Eero-6-Router/dp/B085VM9ZDD>

- [2] Apple. (n.d.). Buy 14-inch MacBook Pro. Apple. Retrieved November 8, 2025, from <https://www.apple.com/shop/buy-mac/macbook-pro/14-inch>

- [3] Apple. (2025, August 21). iCloud+ plans and pricing. Apple Support. <https://support.apple.com/en-us/108047>

- [4] Quicken. (n.d.). Quicken [Brand landing page for subscription information]. Retrieved November 8, 2025, from <https://www.quicken.com/lp/ppc/brand/>

- [5] Google. (2025). Gemini (Version 2.5 Flash) [Large language model]. Response to “Does MacBook Pro come with an automatically updated EDR/Antivirus?” prompt, November 8, 2025. <https://gemini.google.com/>

- [6] Stashi, J.A. (2025). CIDM6341 – Asset inventory & Risk Assessment P1. Unpublished manuscript. College of Business, West Texas A&M University.

- [7] Whitman, M. E., & Mattord, H. J. (2018). Management of information security

[PowerPoint slides]. West Texas A&M University Canvas.
<https://wtamu.instructure.com>